



Lessons Learned Report

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Contents

1	General summary of the project	2
2	Fulfillment of our goals	2
2.1	Follow-up of objectives	2
2.1.1	List of obtained goals	2
3	What we learned (main part)	3
3.1	Good experiences	3
3.1.1	Time Management and Scheduling	3
3.1.2	Testbench	3
3.2	Bad experiences	4
3.2.1	Team Structure and Work Allocation	4
4	Conclusion	4

1. General summary of the project

This project focused on the implementation and verification of a parallel and a bit-serial version of a RISC-V processor. Both processor models needed to be implemented on one FPGA and tested by an advanced testbench implemented on another FPGA, with the CPU and testbench communicating through physical wires. This setup is intended to operate in the extreme environments of Venus, specifically under high-temperature conditions. The work uses KTH's experimental Silicon-on-Insulator technology, which can operate at temperatures approaching Venus's 480°C requirement. However, the test station has a constraint of only 12 pads available, requiring the development of a serialization system and communication protocol.

The project was carried out by dividing the work into two main sub-projects: the testbench development with test set and implementation, and the communication system with serialization and protocol for data transfer. This separation enabled parallel development and allowed team members to focus on different topics and architectures, ultimately achieving better efficiency. The team was organized into two subgroups: three members worked on the testbench (functional testing, CoreMark generation, and integration with RAM and UART), while three members focused on the bit-serial interface (frequency and pin properties, communication protocol specification, and RTL implementation). A significant portion of the work involved learning and adapting to unfamiliar tools, technologies, and design constraints, which strongly influenced both the schedule and working methods.

Regarding planned versus actual outcomes, the project timeline remained largely on track despite several technical challenges. The buffer time allocated during initial planning proved valuable in absorbing unforeseen delays. From a budget perspective, as an academic project no formal budget was involved, though minor unplanned expenses occurred for purchasing power adapters and communication cables between FPGA boards, which team members handled individually. In terms of results, the project successfully delivered a functional dual-FPGA system with a CPU and testbench connected by bit-serial communication. The test for CISC-V and testbench across 2 FPGAs runs successfully with measured performance at 100,198 IPS approaching the 1MHz clock of the CISC-V. The RISC-V has also been tested with the testbench, but only in simulation without the bit-serial communication.

Several significant problems impacted the project timeline and workflow. The initial codebase delivery was delayed, and upon receipt contained various compilation errors and bugs that were assumed to be outside the project scope and already corrected. This required the team to spend considerable time debugging and refactoring code that should have been functional. Additionally, some FPGA boards malfunctioned and needed replacement, and students had to purchase their own adapters and cables for inter-board communication. While these hardware issues were relatively minor and easily resolved, they introduced delays in the integration phase. The Christmas holiday period presented organizational challenges, as not all team members could work together in person. This necessitated a division of tasks between those working remotely and those with physical access to the two FPGA boards required for final integration. Since each member initially had one board, those who traveled left their boards with team members remaining on campus to enable dual-board testing to continue. The holiday period also reduced overall team availability, requiring careful task prioritization and remote workflow coordination. Despite these obstacles, the project maintained progress through adaptive planning, clear task division, and effective cross-team coordination.

2. Fulfillment of our goals

2.1. Follow-up of objectives

The project successfully achieved the majority of its stated technical and business objectives, demonstrating a functional dual-FPGA validation platform despite encountering significant implementation challenges.

2.1.1 List of obtained goals

The following table summarizes the achievement status of all project objectives defined in the initial planning phase:

ID	Objective	Status	Notes
T1	Validation of both processor architectures using a suite of at least 10 assembly-level test programs to verify core ISA functionality	Partially achieved	Both processors validated in simulation; only CISC-V validated on hardware
T2	Execution and verification of a complex benchmark algorithm (CoreMark), with results transmitted to the host PC	Achieved	Successfully executed in simulation for both processors
T3	Demonstration of stable data transfer exceeding 1,000 instructions without corruption between the two FPGAs	Achieved	Verified during CISC-V hardware testing with measured performance of 100,198 IPS
T4	(Optional) Collection of quantitative performance metrics such as maximum clock frequency, benchmark cycle counts, and FPGA resource utilization	Partially achieved	Maximum clock frequency measured; comprehensive resource analysis incomplete
B1	Complete all design, implementation, and validation phases within the planned 16-week schedule	Achieved	All major deadlines met despite technical challenges
B2	Strengthen teamwork, communication, and project management skills	Achieved	Effective collaboration and adaptive planning throughout the project
B3	Deliver a high-quality, fully functional system and comprehensive documentation	Achieved	Functional dual-FPGA system delivered with documentation

Table 1: Summary of project objectives and achievement status

The primary unfulfilled objective was the hardware validation of the RISC-V processor on the dual-FPGA platform. While the RISC-V design passed all functional tests in simulation and was verified to implement the instruction set correctly, integration with the bit-serial communication interface was not completed. This limitation stemmed directly from discrepancies in the initial codebase delivery. The project planning assumed both processor implementations would be provided as consistent black-box modules with identical external interfaces, enabling straightforward integration with the testbench through a common protocol. However, the delivered implementations featured different interface specifications, requiring significant additional integration effort that could not be accommodated within the project timeline after addressing codebase debugging needs.

The successful completion of the CISC-V validation demonstrates that the project methodology is sound and that the remaining objectives are achievable with focused effort on interface standardization and systematic performance characterization.

3. What we learned (main part)

3.1. Good experiences

3.1.1 Time Management and Scheduling

Our project timeline was not average, it was a "slow start" phase followed by high-intensity execution. The initial work was hindered by the steep learning curve of the legacy codebase—specifically the mix of VHDL and ProGram models—and logistical delays in acquiring necessary FPGA hardware. Furthermore, reliance on external feedback for complex toolchains occasionally led to stagnation, highlighting the critical need for proactive contingency planning when guidance is delayed.

Despite these early setbacks, the team successfully adapted by using the mid-term review as a catalyst to overcome inertia and accelerate development. We established a rigorous weekly meeting mechanism to synchronize the disparate timelines of processor implementation and testbench design. Additionally, individual time management played a key role; members compensated for planned absences with periods of intense workload to ensure fairness and adherence to final delivery milestones. This experience taught us that successful time management in experimental projects requires not just scheduling, but the agility to transition from stagnation to crisis management when faced with technological uncertainty.

3.1.2 Testbench

The development of the test platform provided important insights into methodologies for verifying hardware under extreme environmental conditions. A key technical lesson learned was the effectiveness of maintaining a strict "black box" level of abstraction throughout the project. By designing the test environment to interact only through the processor's

top-level instructions and data interfaces, rather than relying on internal microarchitectural signals, we ensured good portability between simulation and the final hardware implementation.

A key decision in our verification strategy was choosing a streamlined version of the CoreMark benchmark as the primary test workload. We found that simple assembly-level sanity checks are often insufficient to detect subtle data path errors. CoreMark was chosen because it provides a realistic combination of integer arithmetic, control flow, and pointer-based data structures, while being small enough for efficient simulation. This choice played a crucial role in the differential verification phase, where we compared simulation logs with the "golden logs" generated by the Spike architecture simulator. Specifically, the benchmark's use of byte and half-word memory operations exposed a critical flaw in the processor's load data path regarding sign extension—an error that simpler test programs failed to trigger.

Also, the transition from simulation to synthesizable platforms highlights the stringent limitations of physical hardware. This process underscores the importance of a design verification environment that separates behavioral transaction logic from the physical transport layer, enabling us to adapt the system from parallel bus simulation to a high-latency bit-serial hardware interface without redesigning the core logic.

3.2. Bad experiences

3.2.1 Team Structure and Work Allocation

Regarding the division of labor, we initially adopted a functional structure that split our six-person team into two equal groups: three members dedicated to the test platform, and three responsible for communication, synthesis, and clock management. While this allocation seemed logical in theory, it proved less efficient in practice. The test platform workload was not constant, leading to periods of overlap and diminishing returns for the three assigned members.

Afterward, we learned that a more flexible structure would have been better. Instead of a rigid 3-3 split, assigning two members to the test platform and make the third as a "mobile" support engineer would have been more effective. This role would have allowed for dynamic resource shifting to assist colleagues on the synthesis and clock generation tasks, who often had to handle high-pressure components in isolation. This experience underscored the importance of flexible team roles over static functional assignments to ensure a balanced workload across all project phases.

4. Conclusion